Industrial Safety Processes

Buildings & Facility Layout

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Content

- General Considerations
- Lighting & Colours
- Emergency Equipment
- Railways
- Facility Maintenance

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General Considerations

Thinking about the staff:

- What will workers do?
- How should workers do it?
- Where should workers do it?
- Why should workers do it?
- What can happen to workers who do it?

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Location & Environment

- Nearby existing or planned operations – compatibility
- Access for staff, vendors,
- Access for goods / supplies / materials
- Storage for goods / supplies / materials
- Permittable locations
- Pollution, waste, run-off, noise, smells, light pollution

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Location & Environment

- Hot or cold weather, seasonal changes
- Common natural disasters
- Utilities – location, cost, reliability
- Emergency response – local, remote
- Railways, waterway access (shipping)

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Common Design & Layout Considerations

- Space for storage, stock, finished product
- Width of aisles for forklift traffic
- Type of lighting and effects on visibility
- Airflow through corridors, open doorways, up shafts
- Separating incompatible activities

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Common Design & Layout Considerations

- Access to items requiring regular maintenance
- Eliminating restricted or confined spaces
- Separating hazardous materials storage
- Employee, supplier access (parking, loading docks)

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Lighting Issues

- More illumination usually results in increased ease, speed and accuracy of accomplishing tasks
- Glare can be an issue:
  - Direct Glare
  - Reflected Glare
  - Discomfort Glare
  - Disability Glare

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More Lighting Issues

- Some types of lighting require additional precautions due to special hazards
  - Metal halide and mercury vapor lamps
- Some types of lighting can change the appearance of standard colors, rendering color-coding efforts ineffective
  - High pressure & low pressure sodium lamps
  - Certain mercury vapor lamps
- See movie Valley of Elah for example of a car color being misidentified due to lighting

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Coloring

- Greatly affected by type of lighting used
- Can affect employee moods, productivity, and safety
- Should be used consistently within facilities
- Following known standards or recognized practices in safety should be followed

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More Color Considerations

- Noise levels
  - Color does not reduce machine noise – but, blues, greens, and neutral colors have a calming effect
- Type of equipment
  - Neutral background, highlight safety colors
- Safety
  - Color schemes, patterns/combos may cause issues

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Typical Safety Colors

- Red – fire, emergency, danger
- Green – first aid, safety
- Yellow – caution
- Orange – dangerous equipment
- Blue – warning
- Black on yellow – radiation, biohazard

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ASME/ANSI A13.1 2015

- Scheme for the Identification of Piping Systems
- Not adopted as a legal requirement
- Often building-specific color schemes are developed but:
  - Buildings change ownership and management
  - It can be very expensive to change piping color in a building

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Coloring by Law

Some colors are controlled by legislation. In Alberta, for example:

RED – not on piping containing flammable or combustible liquids (Canada Fire Code)

YELLOW – required for painting, labeling, or banding natural gas piping (Natural Gas and Propane Installation Code)

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Other Piping Considerations

- Protect natural gas piping into buildings (and inside buildings where necessary) against accidental vehicle damage
- Natural Gas & Propane Installation Code requirements include identifying piping going to specific suites in multi-tenant occupancies

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Emergency Equipment

- Emergency power generators
- Location of emergency fuel supplies
- Types of fire suppression systems
- Exits and stairways
- Specialized detection systems (refrigerant leaks, moisture sensors)
- Backup plans

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Railways

- Do you need access to one?
- Know what you need to do with your railway spur
- Track inspections
- Crossing tracks – vehicles/pedestrians
- Connecting to tanker cars
- Moving cars
- Securing cars – blue flags/lights, derails

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Railway Inspections

- Monthly with at least 20 calendar days between inspections
- Or, before use if used less than once per month
- More often after fire, flood, storm, extreme high or low temperature
- Records must be kept
- Must be trained (standards are vague!)

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Maintaining Maintenance

- The “broken window” theory
- Neat and tidy, well kept, and clean = pride, ownership, and commitment
- One of the best ways of preventing or addressing developing hazards
- Housekeeping is critical
- Slips, trips, and falls are leading causes of injury in most workplaces

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